

OPENSTAX CALCULUS: FUNDAMENTAL THEOREMS OF INTEGRATION

Section: Fundamental Theorem of Calculus

The Fundamental Theorem of Calculus establishes the vital link between differential calculus and integral calculus.

Part 1: If f is continuous on $[a, b]$, then the function $g(x) = \int_a^x f(t) dt$ is continuous on $[a, b]$ and differentiable on (a, b) , and $g'(x) = f(x)$.

Part 2 (Evaluation Theorem): If f is continuous on $[a, b]$ and F is any antiderivative of f , then $\int_a^b f(x) dx = F(b) - F(a)$.

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Section: Integration by Parts Formula

Integration by parts is the integral analog of the product rule for differentiation.

Formula: $\int u \, dv = u \cdot v - \int v \, du$.

Selection rule: Choose u in the priority order ILATE (Inverse trig, Logarithmic, Algebraic, Trigonometric, Exponential). For example, to integrate $x \cdot \cos(x) \, dx$, let $u = x$ (algebraic) and $dv = \cos(x) \, dx$ (trigonometric). Then $du = dx$, $v = \sin(x)$, giving $x \cdot \sin(x) + \cos(x) + C$.